

Matrix Concepts

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1 Matrix Properties

System of Linear Equations: For the system of equations
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots a_{mn}x_n = b_m \end{cases}, Ax$$

= b where $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$ is equivalent. The system can be solved by applying operations to the matrix.

Rank: This is the maximum number of linearly independent rows in a matrix. The rank of a matrix can be found by reducing a matrix and checking for linear independence with the simplified vectors.

Kernel: Also known as the null space, the kernel of a matrix A, $\text{Ker}(A)$, is the vector space of all solutions to the equation $Ax = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$. These are the vectors mapped to the zero vector by the matrix A.

Rank-Nullity Theorem: This theorem states that the dimension of the kernel of a matrix plus its rank is equal to the number of columns in the matrix.

Example: Consider the system of
$$\begin{cases} 3x + 5y + z = 8 \\ 5x - y + 5z = 2 \\ 2x + y + 2z = 5 \end{cases} \quad .$$
 Find the solution to this system by setting up an equivalent matrix equation. Then, find the rank and kernel for this matrix.

2 Eigenvalues/Eigenvectors

Eigenvector: An eigenvector's direction is unchanged when a linear transformation is applied to it. For an eigenvector v of a matrix A , the result of the multiplication Av will have the same direction, but may have a different length. The eigenvector of a matrix A can be found by calculating the solution v to the equation $Av = \lambda v$ or $(A - \lambda I)v = 0$, given an eigenvalue λ .

Eigenvalue: An eigenvalue is the scale of an eigenvector v 's stretch or shrink as the result of the linear transformation A being applied to it. The eigenvalue of a matrix A can be found by calculating the solution λ to the equation $Av = \lambda v$, $(A - \lambda I)v = 0$, given an eigenvector v . Without the eigenvector, the eigenvalues can be found by getting a solution to the equation $\det(A - \lambda I) = 0$.

Example: Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$. Find the eigenvectors and eigenvalues for this matrix.